Clock Arithmetic and some Mysteries of Prime Numbers

Foreword
This article is written for my patient friends to whom I give math riddles that require knowledge of modular arithmetic, or as I often call it *clock arithmetic*. This article serves as an easy reference for the necessary concepts and requires no mathematical background. For this entire article, we will only care about which hour the hour-hand is on; no AM or PM, and no adding half an hour (or anything that isn't an increment of 60 minutes).

Note that any exercise an applications can be skipped; they are just there for fun!

1 Clock Arithmetic: Addition and Multiplication

We all have (at least) two ideas of addition in our heads. The first one is the usual notion of adding quantities[1]: If I have 4 bottles and add 10 bottles, I have 14 bottles. In other words

$$4 + 10 = 14$$

The other notion of addition we often use is when we are calculating what time it is. Consider a 12-hour clock. If it is 4 o'clock, in 5 hours it will be 9 o'clock. If instead we add 10 hours, it will be 2 o'clock. In other words, we just comfortably did

$$4 + 10 = 2$$

in our heads. This is certainly different than the previous equality, but it still makes sense! Using the same symbol "+" may cause confusion, so to distinguish the two types of addition, we shall write $+_{12}$ for "12 hour clock" addition, and keep + for quantity addition. Thus, we can write:

$$4 + 10 = 14 \qquad \text{and} \qquad 4 +_{12} 10 = 2$$

Let us call the addition using + *natural arithmetic* and the addition using $+_{12}$ *clock arithmetic*. As good mathematicians, we should see what similarities and differences these two ideas of additions have.

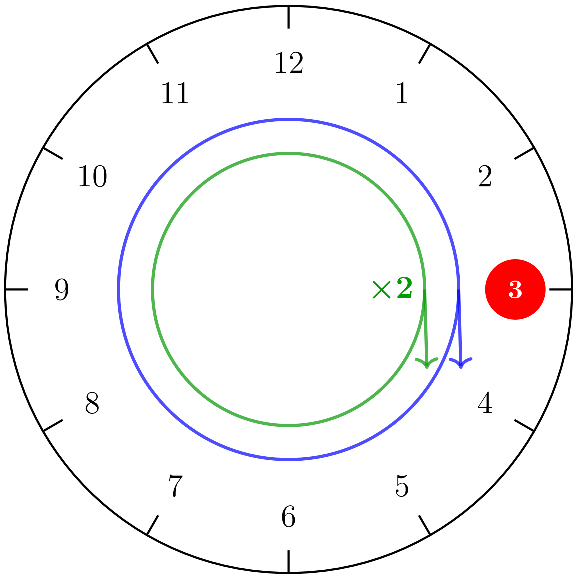
The first thing we should do is add various hours and see what we get. Let us start by picking two random numbers and playing around with them. If it is 3 o'clock, then if I add 5 it will be 8 o'clock. But if I add 17 hours to 3 o'clock, it will *still* be 8 o'clock. More generally, adding any multiple of 12 will not change the result of my addition:

$$3 +_{12} 5 = 8 \qquad 3 +_{12} 17 = 8 \qquad 3 +_{12} 29 = 8 \qquad 3 +_{12} 41 = 8 \cdots$$

So adding $5, (5 + 1 \cdot 12), (5 + 2 \cdot 12), ...$ to 3 gives the same result. Similarly, if I have any number n

[1] A mathematician would say *discrete quantities* to distinguish from *continuous* quantities, which would describe something like water where the amount we add is on a "continuous" spectrum of possible amounts rather than "discrete" indivisible chunks of amounts

between 1 and 12, if I add b, or I add $(b + k \cdot 12)$ for some $k = 1, 2, \ldots$ I get the same result.



$$3 +_{12} 12 = 3$$

$$3 +_{12} 24 = 3$$

Starting position (n = 3)
→ Adding 12 hours
→ Adding 24 hours

+12 hours (1 full rotation)

+24 hours (2 full rotations)

We would like to remember that adding $k \cdot 12$ to any given hour gives the same result. Mathematicians usually use the symbol $\equiv_{12}$ between two such numbers:

$$5 \equiv_{12} 17 \equiv_{12} 29 \equiv_{12} \cdots$$

where we say that 5 is *equivalent* to 17, which is *equivalent* to 29, and so on. Put differently, two numbers $n + a, n + b$ are equivalent if they give you the same time on the clock-face after addition:

$$n +_{12} a = m \qquad \text{and} \qquad n +_{12} b = m$$

Therefore, in clock arithmetic we only need to work with the following numbers:

$$1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12$$

These numbers are *not* equivalent to each other (ex. $3 \not\equiv_{12} 5$). Adding any two of these numbers using $+_{12}$ will give another number in the above list:

$$1 +_{12} 1 = 2 \qquad 11 +_{12} 7 = 6 \qquad 9 +_{12} 12 = 9$$

The last of these three examples is somehow different than the others; it is the only equation where one of the numbers on the left hand side of the addition $+_{12}$ also appears on the right hand side of the equality; that is 9 appears on both sides. In fact, if we try adding any other number to 9, we see that only adding 12 returns 9:

$9 +_{12} 1$	=	10	$9 +_{12} 5$	=	2	$9 +_{12} 9$	=	6
$9 +_{12} 2$	=	11	$9 +_{12} 6$	=	3	$9 +_{12} 10$	=	7
$9 +_{12} 3$	=	12	$9 +_{12} 7$	=	4	$9 +_{12} 11$	=	8
$9 +_{12} 4$	=	1	$9 +_{12} 8$	=	5	$9 +_{12} 12$	=	9

The natural next question to ask is if adding 12 to any number n, not just 9, will return n. Try doing this for any n between 1 and 12 (including 12), and you'll see that adding 12 to any number in clock arithmetic gives back the same number, and 12 is the *only* number that does this!

$$n +_{12} 12 = n$$

Notice that in natural arithmetic, this is exactly the behaviour of 0! That is, for any natural number n:

$$n + 0 = n$$

With this parallel in mind, when we are doing clock arithmetic, we should think of 12 as being the same as 0. Mathematicians would even go so far as re-labeling 12 as 0 so that the 12 numbers of clock arithmetic are:

$$0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11$$

This gives us a good understanding of addition in clock arithmetic: any number n can be re-written as one of the above 12 numbers since n is equivalent to one of these, and then it becomes easy to perform arithmetic.

What about multiplication? Is there a natural way to multiply hours with each other. In natural arithmetic, we know that multiplying n by some natural number m means we are adding n to itself m times. Furthermore, $n \cdot 0 = 0$[2]. Inspired by these properties, let us define multiplication $n \cdot_{12} m$ to mean that we are starting at 12 o'clock (which we should think of as 0 o'clock), and we are moving forward n hours a total of m times.

This matches our intuition on repeated addition. One difference from natural arithmetic and clock arithmetic is that in clock arithmetic there are numbers that are equivalent to each other. We should verify that if we multiply by two equivalent numbers, we ought to get back equivalent results. More precisely, if $n \cdot_{12} m = a$ and $n \cdot_{12} m' = b$, then if $m \equiv_{12} m'$ we want that $a \equiv_{12} b$ (try this out with some numbers!)

Let us start by showing that multiplying by 0 and multiplying by any number equivalent to 0 gives the same result (ex. $n \cdot_{12} 0 = n \cdot_{12} 12 = 0$). Let us first test if this is the case with an example. Let $n = 3$. Certainly if we add no multiples of 3 we stay at 12 o'clock (again, this should be thought of as 0 o'clock). If we add 12 multiples of 3, then we are adding 3 o'clock to itself 12 times. This brings us 36 hours forward, which on a clock-face is 12 o'clock. But we already saw that we should think of 12 as being 0! Hence, we see that adding 3 to itself 12 times will return 0. In fact, if we start with any number n between 0 and 11, adding the number to itself 12 times will always bring us back to the 12 o'clock position! Therefore, 12 really *does* behave like 0 in clock arithmetic! We just showed that multiplying by 12 is like multiplying by 0. To go back to our example using 3:

$$3 \cdot_{12} 12 = 12$$

because 36 hours after 12 o'clock is again 12 o'clock. In our re-labeled system (where 12 is 0) this becomes:

$$3 \cdot_{12} 0 = 0$$

Exercise 1: Multiplying by equivalent numbers

The above shows that if $0 \equiv_{12} m$, then $n \cdot_{12} m = 0$. Show this is true more generally:

$$\text{Let } n \cdot_{12} m = a \text{ and } n \cdot_{12} m' = b. \text{ If } m \equiv_{12} m', \text{ then } a \equiv_{12} b$$

[2] For those who have contested this by asking "what if we let n be infinity", there is this article addressing exactly how to think about infinity and why we shouldn't conclude $0 \cdot \infty = 0$ without taking great care

Exercise 2: Multiplying by other numbers
When multiplying in natural arithmetic we know that $a \cdot b = 0$ if and only if $a = 0$ or $b = 0$. In clock arithmetic, it is now possible that $a \cdot_{12} b = 0$ but $a, b \neq 0$!

Can you think of two non-zero numbers a, b where $a \cdot_{12} b = 0$?

Exercise 3: Combining Addition and Multiplication
Show that for any natural numbers a, b, c

$$a \cdot_{12} (b +_{12} c) = (a \cdot_{12} b) +_{12} (a \cdot_{12} c)$$

Mathematicians call this *distributivity*

Exercise 4: Subtraction
Subtraction works just the same as addition. Convince yourself that:

$$1 -_{12} 2 = 11$$

that is, if it is 1 o'clock, two hours ago it was 11 o'clock. Convince yourself that:

$$a -_{12} 12 = a$$

2 From Clock Arithmetic to Modular Arithmetic

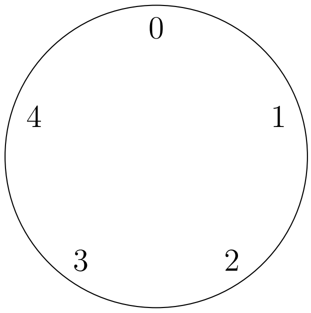
We just showed how we can think of clock arithmetic which has 12 numbers. A mathematician would then naturally ask the following question:

What if we replace 12 with any other natural number[3]?

Let us do exactly this and see what we get. First of all, instead of writing $+_{12}$, we shall now write $+_n$. Let us call the arithmetic for all $+_n$ *modular arithmetic*. With this vocabulary, clock arithmetic is a special case of modular arithmetic. To distinguish the different modular arithmetics, we would usually say we are working *modulo* n. Clock arithmetic would be modular arithmetic modulo 12. When working modulo n, we can imagine a clock face with n positions. For example, if $n = 5$, then

[3]A natural number is of the form $1, 2, 3, ..., n, ...$. We in fact *can* replace it with non-natural numbers, but it took history a long time to understand what this means; this would be introduced in a course in *abstract algebra* (see[4] if you would like my exposition on abstract algebra)

we can think of having a clock with 5 hours:



On this clock, $3 +_5 3 = 1$ and $2 \cdot_5 4 = 3$. When working modulo n, we will always have n numbers:

$$0, 1, 2, ..., n-1$$

Just as for clock arithmetic, we can treat n like 0.

Exercise 5: Zero in Modular Arithmetic

Let a, n be any natural numbers. For a moment, let us not label n as 0. Convince yourself that:

$$a \cdot_n n = n$$

This shows that in modular arithmetic mod n, we can *always* treat n as 0, and hence why we will always write 0 instead of n.

Exercise 6: Special Property when Modulo is Prime

Remember in exercise 2 that in modulo 12, we can find two natural numbers a, b where $a \cdot_{12} b = 0$, but $a, b \neq 0$.

Convince yourself that if n is a prime number (ex. $n = 2, 3, 5, 7, 11, ...$) then

$$a \cdot_n b = 0 \qquad \text{if and only if} \qquad a = 0 \text{ or } b = 0$$

This shows that primes are somehow "special" in that they behave more like natural arithmetic. Mathematicians even have a different word for when n is a prime: modular arithmetic modulo n where n is a prime is called a *finite field*.

3 *Formal Definition of Modular Arithmetic

This section outlines how modular arithmetic is defined formally in mathematics, and benefits the reader who is starting their undergraduate. Let $\mathbb{Z}$ denote the set of integers. Recall an equivalence relation $\sim$ defined on $\mathbb{Z}$ is a subset of the product $\sim \subseteq \mathbb{Z} \times \mathbb{Z}$ satisfying:

1. (reflexivity): $(a, a) \in \sim$ for all $a \in \mathbb{Z}$
2. (symmetry): If $(a, b) \in \sim$, then $(b, a) \in \sim$
3. (transitivity): If $(a, b), (b, c) \in \sim$, then $(a, c) \in \sim$

To simplify notation, we write $a \sim b$ instead of $(a, b) \in \sim$. Recall (or re-prove) that an equivalence relation on $\mathbb{Z}$ partitions $\mathbb{Z}$ (sets whose union is $\mathbb{Z}$ and whose pair-wise intersection is empty); each partition is called an equivalence class.

Define the equivalence relation $\sim_n$ on $\mathbb{Z}$ given by:

$$a \sim_n b \qquad \text{if and only if} \qquad n \big| a - b$$

The collection of equivalence classes will be denoted $\mathbb{Z}/n\mathbb{Z}$. For example, if $n = 12$, then the equivalence classes are:

$$\{[0],\ [1],\ [2],\ [3],\ [4],\ [5],\ [6],\ [7],\ [8],\ [9],\ [10],\ [11]\} = \mathbb{Z}/n\mathbb{Z}$$

If two integers a, b are equivalent modulo n, then $[a] = [b]$. Another very common notation is:

$$a \equiv b \mod n$$

which we denoted earlier as $a \equiv_n b$. The reader should verify that if $a + b = c$ then $a + b \equiv c \mod n$, and similarly if $a \cdot b = c$, then $a \cdot b \equiv c \mod n$. Thus, these equivalence relations preserve the arithmetic properties of $\mathbb{Z}$; such an equivalence relation is called a *congruence relation*. If the reader has seen the axioms of a field, then if $n = p$ is prime, $\mathbb{Z}/p\mathbb{Z}$ satisfies the axioms of a field; otherwise there is always two numbers $[a], [b] \in \mathbb{Z}/n\mathbb{Z}$ such that $[a] \cdot [b] = [ab] = [0]$.

Applications

Now that we have defined modular arithmetic, what can we do with it? The rest of this article will now go over some interesting problems that demonstrate the usefulness of modular arithmetic, as well as reveal some hidden patterns in prime numbers.

4 Solving Equations

The principal reason mathematicians care about modular arithmetic is that it allows us to give a lot more information about equations we are trying to solve. For example, consider $x^2 + x + 1 = 0$. A natural question to ask is whether there is an integer n (numbers of the for $\pm 1, \pm 2, \pm 3, ...$) that solves this equation. In other words, we are looking for an n such that

$$n^2 + n + 1 = 0$$

It may take you a little while to figure out if such a number exists. A natural starting point is to just convince yourself that no small integer can solve it. By plugging in $\pm 1, \pm 2, \pm 3$ we get:

x	Calculation	Result
1	$1^2 + 1 + 1$	3
-1	$(-1)^2 + (-1) + 1$	1
2	$2^2 + 2 + 1$	7
-2	$(-2)^2 + (-2) + 1$	3
3	$3^2 + 3 + 1$	13
-3	$(-3)^2 + (-3) + 1$	7
$\vdots$	$\vdots$	$\vdots$

Doing this for the first 100 integers and their negatives would show you that the equation never equals 0, however there is no guarantee that you will *never* find a big enough number n such that $n^2 + n + 1 = 0$. Using modular arithmetic, it is in fact straightforward to show that *no* solutions exist.

To show this, let's assume that there *does* exist an n such that $n^2 + n + 1 = 0$[4]. Then since $n^2 + n + 1 = 0$, certainly $n^2 + n + 1 \equiv_2 0$: if we let $n^2 + n + 1 = m$, then $m \equiv_2 0$ means that m is a multiple of 2. But 0 is certainly a multiple of 2 since $0 \cdot 2 = 0$. Said differently, As 0 is an even number, $n^2 + n + 1$ must be even. What we will show is that it is *impossible* to find an integer n such that $n^2 + n + 1 \equiv_2 0$; in other words this equation has no solutions modulo 2.

First of all, for any integer n:

$$n \equiv_2 0 \qquad \text{or} \qquad n \equiv_2 1$$

[4]This is a usual trick called *proof by contradiction.* This idea is to assume the statement we want to prove false, is true, and show this creates a contradiction

More concretely, any integer must either be even or odd. For example, if $n = -177$, then:

$$-177 = 1 + 2 \cdot (-89) \equiv_2 1$$

(look at exercise 4 if the subtraction is confusing). Thus, every natural number is equivalent to either 1 (if it is odd) or 0 (if it is even). Armed with the possible values of n modulo 2, we get one of two possible equations:

$$(1 \cdot_2 1) +_2 1 +_2 1 = 0 \qquad \text{or} \qquad (0 \cdot_2 0) +_2 0 +_2 1 = 0$$

But now, if we compute the left hand side of the first equation we get

$$1 +_2 1 +_2 1 = 0 +_2 1 = 1$$

which is *not* equal to the right hand side (i.e. $1 \neq 0$ and $1 \not\equiv_2 0$). The second equation is even more obvious since $(0 \cdot_2 0) +_2 0 +_2 1 = 1$, which is again not equal to 0. Concretely, if n is odd, then n^2 is odd, and adding three odd numbers is always odd. If n is even, then n^2 is even, and adding two even numbers and an odd number is odd.

So we arrived at a contradiction, because $0 \not\equiv_2 1$. What this tells us is that we made a wrong assumption earlier in the proof; and indeed this started by assuming that there exists an integer n such that $n^2 + n + 1 = 0$. Thus, it is *impossible* that such an integer exists! Notice how easy this was to prove: we only had to check *two* possible numbers in modular arithmetic modulo 2 as compared to the infinite possibilities when trying to solve the equation with natural numbers!

This "reduction to finitely many cases" is one of the *key* properties of modular arithmetic that mathematicians abuse to no end.

Exercise 7: No Integer Solution I

Show that $2x^2 + x + 1 = 0$ has no integer solutions (hint: consider modular arithmetic modulo 3)

Exercise 8: No Integer Solution II

This question is *very* hard, and is here to show that the modulo technique is insufficient of a tool for finding integer solutions to *any* equations. Consider the polynomial

$$3x^3 + 4y^3 + 5z^5 = 0$$

Show that this polynomial has *no* integer solutions, but it *does* have a solution modulo n for every natural number n! Due to this polynomial, a mathematician would say it is a *necessary* but not a *sufficient* condition for a polynomial to have a solution if it has a solution modulo any n

It is equations like these that prompted a large wave of development in number theory known in the 21st century as the theory of *elliptic curves*. The theory of elliptic curves it is certainly a graduate level topic, but the curious reader may checkout [1] for my exposition on the topic.

5 Application: Patterns in Prime Numbers

Let us now show that modular arithmetic reveals hidden patterns in primes. Consider the equation $x^2 = p$ for some prime number p. Then certainly no natural number solves this equation, because no natural number can be the square-root of a prime[5]. However, this question is more interesting if we instead consider it in *modular arithmetic* modulo n.

[5] This is because the square-root of a prime number is *irrational*; see [3] for a friendly introduction to this notion

Let us first define precisely what it means for a number to be a square root. A number a is the square root of a number m if $a^2 = m$. For example, 4 is the square-root of 16 because $4^2 = 16$. If m is a prime, then there is no *natural number* a such that $a^2 = p$.

So in natural arithmetic, square-roots of a prime do not exist. Let us see if we can find a square-root in modular arithmetic in some modulo n. Let us demonstrate this with an example. Consider $p = 3$. Let us see if in modulo n, we can find a number m such that $m \cdot_n m = p$ (i.e. if p has a square-root modulo n). The following is a table where n goes over the first few natural numbers and finds all possibles squares:

n	0^2	1^2	2^2	3^2	4^2	5^2	6^2	7^2	8^2	9^2	10^2
2	0	1									
3	0	1	1								
4	0	1	0	1							
5	0	1	4	4	1						
6	0	1	4	**3**	4	1					
7	0	1	4	2	2	4	1				
8	0	1	4	1	0	1	4	1			
9	0	1	4	0	7	7	0	4	1		
10	0	1	4	9	6	5	6	9	4	1	
11	0	1	4	9	5	**3**	**3**	5	9	4	1

Let us focus on the cases where n is a prime number. Notice that $5 \cdot_{11} 5 = 3$, since $5 \cdot 5 = 25 = 3 + 2 \cdot 11$. Therefore, in modulo 11, we have that:

$$\sqrt{3} = 5 \text{ (in modulo 11)}$$

Hence, there *does* exist a square-root of 3 in modulo 11, and there *doesn't* appear to be a square-root of 3 modulo 5 or 7 (among many other primes). But why 11? Why was the prime 3 somehow linked to the prime 11? The reader can check that there are in fact other primes in which 3 also has a square-root; so how can we determine for which primes 3 has a square-root?

This is a rather fascinating result: some primes have square-roots in modulo p, while other don't, in other words primes somehow have *relations* between each other! To be precise, two primes are tied to each other if one prime has a square-root when doing modular arithmetic modulo the other prime! In the 19th century, Euler and Gauss had the exact same realization, and did thousands of such computations looking for patterns. Gauss provided a rigorous proof explaining the behaviour in a theorem that is now called *quadratic reciprocity*. It is one of the theorem with the most distinct proofs in mathematics (currently having over 200 different proofs known proofs). Presenting one of these proofs would require some more math than this article covers (all the details are in [4], and requires a good deal of build-up before being proven in chapter 22.4). However, I strongly encourage the reader play around with this; after a few hours, the reader will most likely stumble upon the same pattern that many other mathematicians who worked on the problem (including Euler) have discovered!

Exercise 9: Two solutions modulo 11?

Looking at the row for modulo 11, it is clear that there are in fact two numbers that are the square-root of 3, in particular there is also $6 \cdot 6 = 36 = 33 + 3 = (11 \cdot 3) + 3$. This is because in modulo 11, $6 \equiv_{11} -5$, showing this is in fact the same solution times -1. Show that:

$$6 \cdot_{11} 6 = (-5) \cdot_{11} (-5) = (-1 \cdot_{11} -1) \cdot_{11} (5 \cdot_{11} 5) = (5 \cdot_{11} 5)$$

6 Final Mystery

The problem in the above section turned out to be one of the cornerstones of number theory, as it somehow connects different prime numbers to each other via equations; that is, prime numbers are not completely "random" but can have subtle connections that are not obvious at first glance!

In the same vein, here is the final problem we shall present:

> Are there 0, finitely many, or infinitely many primes p such that
>
> $$p \equiv_5 3$$
>
> In other words, how many primes are there for which p o'clock in a 5 hour clock will be the same time as 3 o'clock.
>
> The numbers 3 and 5 can be replaced with any numbers a, b where $\gcd(a, b) = 1$ (i.e. they have no common factors).

Doing the first few primes, we see that the answer is at least finitely many:

prime	2	3	5	7	11	13	17	19	23	29	31	37	41	43	47	53	59	$\cdots$
$p \equiv_5 m$	2	3	0	2	1	3	2	4	3	4	1	2	1	3	2	3	4	$\cdots$

Table 1: First 17 primes and their values modulo 5

This question was pondered by Dirichlet in 1837. Essentially, it is asking if there is a meaningful way in which prime numbers can be grouped together. A mathematician would say that this is asking how the primes numbers are *distributed*. Just like the above problem, I strongly encourage the reader to think about this problem for a couple of hours and just see what they can come up with. The result has been proven, and a full proof requires more math than what can be covered in an expository article like this (for completeness, a proof can be found in [2, chapter 3.4]). I hope the reader will still ponder the problem just as many mathematicians have done before, for it is by trying to solve such hard questions that mathematicians discover novel insights.

Exercise 10: Common Factors
Consider the same question, but instead let a, b have common factors. Will there always be infinitely many or finitely many primes p such that $p \equiv_b a$? This question is solvable; can the reader provide a proof?

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